DSC-002: Mathematical Methods

(Linear Algebra)

Lecture 5: Spectral Theory

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Notion & Definition: Eigen values, Eigen vectors

Amount of scaling

vector whose direction is preserved

Let $T : V \to V$ be a LT. A scalar λ is an **eigenvalue** of T if there exists a non-zero vector $v \in V$ s.t. $T(v) = \lambda v$. $\Rightarrow \lambda$ = Eigenvalue

Where v is an **eigenvector** of T, corresponding to the eigenvalue λ.

All the cv will follow same eigenspace

Note (1) If v is an eigenvector corresponding to λ, then so is cv, for all scalars c, because $T(cv) = cT(v) = c\lambda v = \lambda(cv)$. eigenvector

Moreover, if v, v' are eigenvectors corresponding to the same eigenvalue λ, then for scalars c_1, c_2 we have -

$$T(c_1 v + c_2 v') = c_1 T(v) + c_2 T(v')$$
$$= c_1 \lambda v + c_2 \lambda v' = \lambda(c_1 v + c_2 v').$$

linear combination is also eigenvalue

So $(c_1 v + c_2 v')$ is also an eigenvector corresponding to λ. That is, the set of all eigenvectors corresp. to a particular eigenvalue is a **subspace**. It (along with the zero vector) is called the **eigenspace** corresponding to λ.

Notions

Note (2) For a typical $(n \times n)$ matrix A and a typical $(n \times 1)$ vector v, the vector Av is some point in $\mathbb{R}^n$ (or $\mathbb{C}^n$, if A is a complex matrix).

In the eigenvalue problem, we are looking for vectors v such that Av lies on the same line through the origin as v itself.

Example

(1). A matrix for which *all* non-zero vectors are eigenvectors: $I_{n\times n}$. For all non-zero $x \in \mathbb{R}^n$, $Ix = 1x$; all such x are eigenvectors corresponding to the eigenvalue 1.

(2) Using the appropriate rotation matrix, rotate the earth through say 90^o. Typically, vectors $x \in \mathbb{R}^3$ will change direction on rotation. But vectors lying on the *axis of rotation* will not change direction. For such vectors v, $Av = 1v$.

The set of *all eigenvalues* of an LT T is called the **Spectrum** of T, and denoted $\sigma(T)$.

Characteristic Equation

Let $T(v) = \lambda v$, then $(T - \lambda I)(v) = 0$, i.e., e-vector v corresponding to the e-value λ comprise the *Nullspace* of the LT $(T - \lambda I)$ (along with zero vector). The nullspace $N(T - \lambda I)$ is called the **eigenspace** w.r.t. λ.

- For there to be a non-zero solution v to $(T - \lambda I)(v) = 0$, we need $Nullity(T - \lambda I) > 0$, or $Rank(T - \lambda I) < dim(V)$.
- Let the matrix $A_{n \times n}$ represent T. So we have $Rank(A - \lambda I) < n$, which implies the determinant $det(A - \lambda I) = 0$. $det(A - \lambda I)$ is just

↳ characteristic equation.

$$det \begin{pmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ \vdots & \dots & \ddots & \vdots \\ a_{n1} & \dots & \dots & a_{nn} - \lambda \end{pmatrix}$$

polynomial form

So $det(A - \lambda I) = 0$ is n^{th}-degree polynomial and can be written as

$$f(\lambda) = |(A - \lambda I)| = (-\lambda)^n + b_{n-1}(-\lambda)^{n-1} + \dots + b_1(-\lambda)^1 + b_0 \quad \dots (1)$$

or in factor form,

$$f(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\dots(\lambda_n - \lambda) \quad \dots . (2)$$

↓ multiplication form

Characteristic Equation contd...

Sum of eigenvalues = Trace.
Product of eigenvalues = Determinant

- **Note:** So $\det(A - \lambda I) = 0$ has n solutions (roots), possibly *repeated*, and possibly *complex*.
- **Note:** If λ is a complex root of the equation, its conjugate $\bar{\lambda}$ is also a complex root.

- Comparison of Eqn 1 and Eqn 2 on the previous slide gives us -

$$b_{n-1} = \sum_{i=1}^{n} \lambda_i = \lambda_1 + \lambda_2 + \cdots + \lambda_n$$

Trace(A)

$$b_{n-2} = \sum_{j>i}^{n} \lambda_i \lambda_j = \lambda_1\lambda_2 + \cdots + \lambda_1\lambda_n + \lambda_2\lambda_3 + \cdots + \lambda_2\lambda_n + \cdots + \lambda_{n-1}\lambda_n$$

sum of all two two pairs

$$b_{n-r} = \sum_{k>j>i}^{n} \lambda_i\lambda_j\lambda_k \quad \text{[each term is a product of } r \text{ such } \lambda_i\text{]}$$

$$\vdots$$

$$b_0 = \lambda_1\lambda_2\lambda_3\cdots\lambda_n$$ - multiplication of all eigen values.

Characteristic Equation contd...

The characteristic polynomial for a second-order matrix $A = ||a_{ij}||$ is

$$f(\lambda) = \begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21}$$
$$= (-\lambda)^2 + (a_{11} + a_{22})(-\lambda) + (a_{11}a_{22} - a_{12}a_{21})$$
$$= (-\lambda)^2 + \quad b_1(-\lambda) + b_0$$

Hence, $b_1 = a_{11} + a_{22}$ (i.e. Trace of A), $b_0 = a_{11}a_{22} - a_{12}a_{21} = |A|$

↳ determinant of matrix

Two roots of $f(\lambda) = 0$ are:

$$\lambda = \tfrac{1}{2}\{(a_{11} + a_{22}) \pm [(a_{11} + a_{22})^2 - 4|A|]^{1/2}\}$$

if λ_1 and λ_2 are the two roots, then:

$\lambda_1\lambda_2 = |A| = b_0$ and $\lambda_1 + \lambda_2 = a_{11} + a_{22} = b_1$

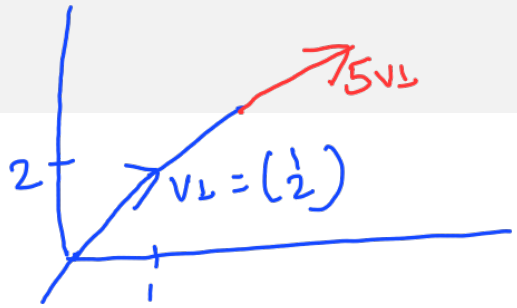
↳ determinant ↳ trace of b1

Recall: A quadratic eqn in λ can be as: $\lambda^2 - \ trace(A)\lambda + \ det(A) = 0.$

Note: The sum of the diagonal elements of a square matrix A (called **trace(A)**) equals the sum of eigen values $\sum_{i=1}^{n} \lambda_i$.

And $det(A) = \Pi_{i=1}^{n}\lambda_i$, i.e. the product of eigen values.

Example 1: Distinct eigen values

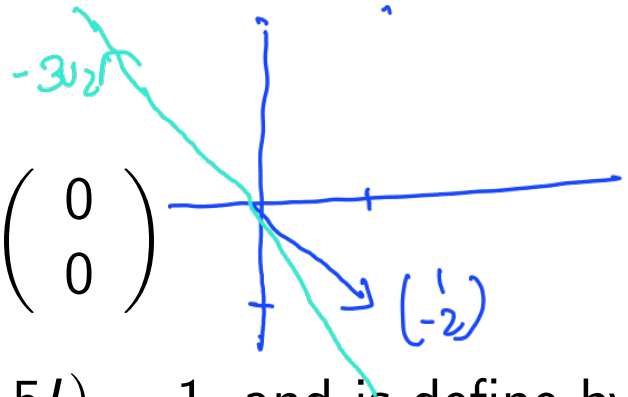


Example 1. Let $A_{2\times 2}$ have rows $(1,2)$ and $(8,1)$.

Thus $det(A-\lambda I)=\lambda^2-2\lambda-15=0$ has solutions 5 and -3.

↳ characteristic eqn

So, eigenvectors for $\lambda_1=5$:



$$\begin{pmatrix} 1-5 & 2 \\ 8 & 1-5 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The eigenspace is 1-dimensional, since $Rank(A-5I)=1$, and is define by $\{(x_1,x_2)\mid -4x_1+2x_2=0\}$. Eqn of the null space

If we choose $x_1=1$, we solve and get $x_2=2$, so $x=(1,2)^T$ is an eigenvector corresponding to $\lambda=5$.

Similarly, $(1,-2)^T$ is an eigenvector corresponding to $\lambda_2=-3$.

λ_1 and λ_2 are spanning the eigenspace.

Example 2: Complex eigen value and conjugate

Example 2: $A_{2\times 2}$ has rows $(1,2)$ and $(-2,1)$.

$\lambda = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ $a=5$ $b=-2$ $c=5$

$det(A-\lambda I) = \lambda^2 - 2\lambda + 5 = 0$ implies the roots equal $\frac{2\pm\sqrt{4-20}}{2}$.

So $\lambda_1 = 1+2i$ and $\lambda_2 = \overline{\lambda_1}$.

Eigenvector corresponding to λ_1:

$$(A-(1+2i)I)x = \begin{pmatrix} -2i & 2 \\ -2 & -2i \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$Rank(A-\lambda_1 I) = 1$ (the 2nd row is i times the first).

Solving $-2ix_1 + 2x_2 = 0$, and by setting $x_1 = 1$, we get $x_2 = i$.

So $(1,i)^T$ is an eigenvector corresponding to λ_1.

The conjugate of this vector, $(1,-i)^T$ is therefore an eigenvector corresponding to the conjugate root λ_2. **This is a general result.**

Q: Is there orthogonality between a vector and its conjugate?

Similar Matrices, Identical Spectrum

Can we replace a linear transformation T with a matrix A or even another matrix B representing it? Indeed all matrices representing T in different bases are similar to each other. If A and B are 2 such matrices, then $A = SBS^{-1}$ for some invertible (change of basis) matrix S.

If there exists a nonsingular matrix S such that $A = SBS^{-1}$ (or, $B = S^{-1}AS$), then the square matrices A and B are said to be **Similar**.

Similar matrices have an identical spectrum - means - similar matrices have the same eigenvalues

Lemma: Similar matrices have an identical spectrum. collection of all λ's

- spectrum will remain same
- Eigenvalue will be transformed

Proof: Let $Av = \lambda v$, so λ is an eigenvalue of A.
If $A = SBS^{-1}$, then this implies $SBS^{-1}v = \lambda v$.
Premultiplying both sides by S^{-1}, we get:

$$S^{-1}SBS^{-1}v = S^{-1}\lambda v \implies B\underbrace{S^{-1}v} = \lambda\underbrace{S^{-1}v}.$$

So λ is an eigenvalue of B (and $S^{-1}v$ is the corresponding eigenvector).

Theorems: Real, Symmetric Matrix ($A = A'$)

If A is a real, symmetric matrix in $Ax = \lambda x$, then e-values of A are real.

Assume, λ is a real /complex eigen value of real symmetric matrix A.
Then $\exists$ at least one eigen vector (may have complex component) s.t.

$$Ax = \lambda x \qquad \cdots (1)$$

Taking the complex conjugate of (1), we obtain

$$A\bar{x} = \bar{\lambda}.\bar{x} \qquad \cdots (2)$$

Note, $\bar{x} = [\overline{x_1}, ..., \overline{x_n}]^T$ and $\bar{A} = A$ *(since A is real).*

Pre-multiplying eqn (1) by x^* $[:= (\bar{x})']$, and eqn (2) by x', and subtracting, we have:

$\lambda = \bar{\lambda} = 0$ - Real = conjugate

left hand side becomes zero

$$\Leftarrow x^*Ax - x'A\bar{x} = (\lambda - \bar{\lambda})x'\bar{x}, \qquad \cdots (3)$$

Since $x^*x = \bar{x}^T x = \sum \bar{x}_i x_i$ and $x'\bar{x} = x^T\bar{x} = \sum x_i \bar{x}_i$, these are equal (scalar multiplication commutes). Also recall that $A' = A$. □

Proof continues ...

Proof

Now, $x'A\overline{x}$ is a number, and the transpose of a number (matrix of a single element) is a number.

Thus $x'A\overline{x} = (x'A \cdot \overline{x})' = (\overline{x})'Ax = x^*Ax.$ (since $A' = A$)

Thus, LHS of (3)=0.

In addition, $x'\overline{x} = \sum x_i\overline{x_i}$ is real and positive since $x \neq 0$.

Consequently, $\lambda = \overline{\lambda}$, so λ is real
($\because$ if $\lambda = a + bi$ and $\overline{\lambda} = a - bi$ and $\lambda = \overline{\lambda} \Rightarrow b = 0$ is only possibility).

Real, symmetric matrix has real eigen-values. □

Theorems: Real, Symmetric Matrix ($A = A'$)

For a real and symmetric matrix, the eigen-vectors corresponding to different eigen-values are orthogonal; i.e., if x_i, x_j are eigen-vectors corresponding to eigenvalues λ_i and λ_j respectively, then $x_i' \cdot x_j = 0$.

λ_i - eigen value, x_i - eigenvector
x_j - eigen vector

Proof

By assumption, $A = A^T$, $Ax_i = \lambda_i x_i$ and $Ax_j = \lambda_j x_j$.

Thus, $x_j' Ax_i = \lambda_i x_j' x_i$ and $x_i' Ax_j = \lambda_j x_i' x_j$.

Note, by symmetricity property: $x_j' Ax_i = x_i' Ax_j$.

Subtracting, we have $(\lambda_i - \lambda_j) x_j' x_i = 0$, or $x_j' x_i = 0$, since $\lambda_i \neq \lambda_j$. □

Note: If $x \neq 0$ is an eigen-vector of A, then any scalar multiple of x is also an eigen-vector of A, corresponding to the same eigen-value.

Orthonormal Eigen Vectors

- Normalize and orthnogonal give us orthonormal vectors.

Note: For a real symmetric matrix, there exists an **orthonormal basis** of eigenvectors. Conveniently, we often normalize each eigenvector to have **unit length** and denote them by $\mathbf{u_i}$, and are said to be normalized.

A set of two or more normalized $\mathbf{u}_i$s corresponding to different eigen-values λ_i of A satisfies a condition $\mathbf{u_i'}.\mathbf{u_j} = 0$ and are called **orthonormal**.

Example 3: Orthonormal eigen vectors

Find the eigen values and eigen vectors of $A = \begin{vmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 1 \end{vmatrix}$.

The characteristic equation is:

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & \sqrt{2} \\ \sqrt{2} & 1-\lambda \end{vmatrix} = \lambda^2 - 3\lambda = 0 \quad \Rightarrow \lambda(\lambda-3) = 0$$

thus the eigenvalues are $\lambda_1 = 0$ and $\lambda_2 = 3$.

To determine eigen vectors corresponding λ_i , we must solve the set of homogenous equation $[A - \lambda I]x = 0$.

Let us take $\lambda_1 = 0$ first, then the set of equations becomes –

$$2x_1 + \sqrt{2}x_2 = 0 \qquad \text{and} \qquad \sqrt{2}x_1 + x_2 = 0$$

Implies, $x_1 = -\frac{1}{\sqrt{2}}x_2$

If we wish to find an eigen vector of **unit length**, we must **require** that $x_1^2 + x_2^2 = 1$. Thus $\frac{3}{2}x_2^2 = 1$; choosing the positive square root, we obtain -

$$x_2 = \sqrt{\frac{2}{3}} \qquad \text{and} \qquad x_1 = -\frac{1}{\sqrt{3}}.$$

And the eigenvector of unit length corresponding to λ_1 be:

$$u_1 = [-\frac{1}{\sqrt{3}}, \sqrt{\frac{2}{3}}]^T$$

Example 3: contd...

Note that u_1 is not completely specified by the requirement of $|u_1| = 1$. The vector $-u_1$ is also an eigen vector of length 1. However u_1 and -u_1 are not linearly independent. Only one linearly independent vector corresponds to λ_1.

For $\lambda_2 = 3$, the set of equations becomes:

$$-x_1 + \sqrt{2}x_2 = 0 \qquad \text{and} \qquad \sqrt{2}x_1 - 2x_2 = 0$$

implies, $x_1 = \sqrt{2}x_2$.

If $x_1^2 + x_2^2 = 1$ then $3x_2^2 = 1$; taking positive square root we get:

$$x_2 = \frac{1}{\sqrt{3}}\ , \qquad x_1 = \sqrt{\frac{2}{3}}$$

So $u_2 = [\sqrt{\frac{2}{3}}, \frac{1}{\sqrt{3}}]^T$.

Again there is only one linearly independent eigen vector that corresponds to λ_2; it can easily be checked that $u_1^{'} \cdot u_2 = 0$, in agreement with the theory, i.e. orthonormal vectors are orthogonal.

Example 4: Repeated Eigenvalues

Consider $A = \begin{pmatrix} -5 & -6 & 3 \\ 3 & 4 & -3 \\ 0 & 0 & -2 \end{pmatrix}$,

So characteristic polynomial: $X^3 + 3X^2 - 4 = 0 \Rightarrow$ Three roots.

Eigenvalues: $\lambda_1 = 1$ & $\lambda_2, \lambda_3 = -2$ (i.e. one e-value is repeated twice!!)

Eigenvectors: $\left\{ \begin{matrix} -1 \\ 1 \\ 0 \end{matrix} \right\} \leftrightarrow \lambda_1 = 1$; $\left\{ \begin{matrix} -2 & 1 \\ 1 & , \ 0 \\ 0 & 1 \end{matrix} \right\} \leftrightarrow \lambda_2, \lambda_3 = -2$

How to produce 2 different e-vectors corresponding to a repeated e-value?

If you put $\lambda_2 = -2$ in the characteristic equation $[A - \lambda_2 I]$ then -

$$\begin{pmatrix} -5+2 & -6 & 3 \\ 3 & 4+2 & -3 \\ 0 & 0 & -2+2 \end{pmatrix} = \begin{pmatrix} -3 & -6 & 3 \\ 3 & 6 & -3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

The exprsn above will give you one eigenvector $(1, 0, 1)^T$. **How?** Rank?

What about the other?

$-3x_1 - 6x_2 + 3x_3 = 0 \rightarrow \left\{ \begin{matrix} 1 \\ 0 \end{matrix} \right\}$

Example 4: Repeated Eigenvalues ... contd ...

RREF

Let's perform some row reductions on the matrix for $[A - \lambda_2 I]$ to obtain pivots. (Note that 3 Columns are dependent, so expect 1 pivot only)

$$\begin{pmatrix} -3 & -6 & 3 \\ 3 & 6 & -3 \\ 0 & 0 & 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} \boxed{1} & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Now, we have to solve: $\begin{pmatrix} \boxed{1} & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

Here x_2, x_3 are free variables, and set their values s and t, respectively. Then solve for $x_1 = -2x_2 + x_3$, $x_2 = s$, $x_3 = t$ and put it in vector form -

$$\begin{pmatrix} -2s + t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

So, $v_2 = (-2, 1, 0)^T$ and $v_3 = (1, 0, 1)^T$. Check if they are L.I.

NOTE: Multiplicity of an eigenvalue: dimension of eigenspace $\ker(A - \lambda I)$ is called **geometric multiplicity** of the eigenvalue λ.

⇒ equal to dimension of Nullspace.

Diagonal Form of Matrix

Th

Suppose $A_{n\times n}$ has 'n' LI eigenvectors. Then $S^{-1}AS = D$, where the columns of S are the 'n' eigenvectors (from $[A-\lambda I]$) and D is the diagonal matrix whose diagonals are the corresponding eigenvalues.

Proof

Note $S = (\mathbf{v_1}, ..., \mathbf{v_n})$ is invertible (eigenvectors written as columns). Then $AS = A(\mathbf{v_1}\ \mathbf{v_2}\ ...\ \mathbf{v_n}) = (A\mathbf{v_1}\ A\mathbf{v_2}\ ...\ A\mathbf{v_n}) = (\lambda_1\mathbf{v_1}\ \lambda_2\mathbf{v_2}\ ...\ \lambda_n\mathbf{v_n})$

$$\text{i.e. } AS = \begin{pmatrix} \vdots & \vdots & \ddots & \vdots \\ v_1 & v_2 & \dots & v_n \\ \vdots & \vdots & \ddots & \vdots \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix} = SD$$

Premultiplying both sides of $AS = SD$ by S^{-1}, $S^{-1}AS = D$. □

When can we Diagonalize?

Either λ are distinct and corresponding eigenvectors are LI] conditions for diagonal

Suppose $A_{n\times n}$ has 'n' distinct (real) eigenvalues $\lambda_1, ..., \lambda_n$. Then a set of eigenvectors $v_1, ..., v_n$ is LI (where, each v_i corresponds to λ_i).

Proof. (By induction). The statement is trivially true if $n = 1$.
Suppose it's true for $n = 1, 2, ..., k-1$; i.e. if $\lambda_1, ..., \lambda_{k-1}$ are distinct and $v_1, ..., v_{k-1}$ are corresponding eigenvectors, then they are L.I.

Suppose λ_k is an eigenvalue distinct from $\lambda_1, ..., \lambda_{k-1}$ and v_k is a corresponding eigenvector.
For some scalars $c_1, \ldots, c_k$, suppose that:

$$c_1 v_1 + \ldots + c_k v_k = 0 \qquad (\mathbf{1}).$$

So, applying the transformation on (1), we get –

$$A(c_1 v_1 + ... + c_k v_k) = A \cdot 0 = 0.$$

$$\Rightarrow \quad c_1 A v_1 + ... + c_k A v_k = 0 \qquad \text{[by linearity of } A\text{]}$$

$$\Rightarrow \quad c_1 \lambda_1 v_1 + ... + c_k \lambda_k v_k = 0 \qquad (\mathbf{2}).$$

When can we Diagonalize?

Proof continues...

Subtracting $\lambda_k \times$Eq(1) from Eq(2) eliminates $c_k \lambda_k v_k$, giving:

coefficients coefficients

$$c_1(\lambda_1 - \lambda_k)v_1 + \ldots + c_{k-1}(\lambda_{k-1} - \lambda_k)v_{k-1} = 0.$$

By linear independence of $\{v_1, \ldots, v_{k-1}\}$, we get –

$$c_i(\lambda_i - \lambda_k) = 0, \quad \forall i = 1, .., k-1.$$

Since $\lambda_i \neq \lambda_k$, we have: $c_i = 0, \quad \forall i = 1, \ldots, k-1$.
Plugging c_i's in Eq(1), we get $c_k = 0$ also, so that $\{v_1, .., v_k\}$ are linearly independent.

Matrix are all eigenvalues in diagonal position.

Why Diagonalize?

(**1**) Interpretation. If matrix A is diagonalizable, then $A = SDS^{-1}$.
That is A and D are **similar**.
Specifically, suppose $x = (x_1, ..., x_n)^T$ is the coordinate vector written in terms of the standard basis.
Suppose the LI eigenvectors $\{v_1, ..., v_n\}$ are the new basis.
If $S_{n\times n} = (\mathbf{v_1}\ \mathbf{v_2}\ ...\ \mathbf{v_n})$, then w.r.t. the new basis, we know that x is represented as $y = S^{-1}x$. Now, the matrix A carries x to Ax.
In terms of the new basis, Ax is represented as $S^{-1}Ax$.
Since $x = Sy$, $S^{-1}Ax = S^{-1}ASy = Dy$.
So wrt the new basis, the action of A on x becomes the easier to visualize action of D on y.
D just stretches the coordinates of y by possibly different amounts.

(**2**) Uses - including more general uses of finding the spectrum $\sigma(A)$:
In solving and interpreting solutions to differential equations; also in some specification testing in econometrics, interpreting quadratic forms, frequency domain time series analysis etc.

Why Diagonalize? ...contd...

Trail ex:- $A \to A^{100}$; $SDS^{-1} \to (SDS^{-1})^{100} \to SD^{100}S^{-1}$; $S^{100} \to S$; $(S^{-1})^{100} \to S$

(**3**) If A is real, symmetric and full-ranked, then we can find a set of eigenvectors that are LI as well as orthogonal.
Then the matrix S of eigenvectors is called an orthogonal matrix.

These are generally rotation matrices;
Thus, $A = S\Lambda S^{-1}$ means the action of A is a composition of a rotation followed by a stretch followed by another rotation.

Examples of such matrices:
(i) the (X^TX) or (X^*X) matrix discussed in the least squares application;
(ii) the Hessian D^2f of a function $f : \mathbb{R}^n \to \mathbb{R}$.

Examples

Continuing from the previous example with the matrix $A = \begin{vmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 1 \end{vmatrix}$

And the eigenvectors of unit length corresponding to λ_1 and λ_2 are -

orthogonal

$$\begin{cases} u_1 = [-\frac{1}{\sqrt{3}}, \sqrt{\frac{2}{3}}] \\ u_2 = [\sqrt{\frac{2}{3}}, \frac{1}{\sqrt{3}}] \end{cases}$$

$$S = (u_1, u_2) = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 & \sqrt{2} \\ \sqrt{2} & 1 \end{bmatrix}$$

$$S'AS = \frac{1}{3} \begin{bmatrix} -1 & \sqrt{2} \\ \sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} -1 & \sqrt{2} \\ \sqrt{2} & 1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} 0 & 3\sqrt{2} \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

.

Note: For matrix with orthonormal vectors: $S^{-1} = S'$. You can verify this claim.